

AI DEPLOYMENT CONSTRAINTS

EEG-Based Clinical Decision Support Tool

Healthcare AI Deployment Readiness Assessment

Author: Bhumi Nagar | Domain: Healthcare AI / Clinical Decision Support | Document Type: Deployment Constraints

CURRENT DEPLOYMENT STATUS: NOT DEPLOYMENT READY

The current project is a proof-of-concept feasibility study. The constraints documented here must be resolved before any clinical deployment can be considered.

1. Purpose

This document identifies and analyzes the deployment constraints that would govern the transition of the EEG-based clinical decision support system from a proof-of-concept feasibility study to a real-world clinical environment.

These constraints are not blockers to the feasibility study itself — they are the conditions that must be resolved before any clinical deployment can be considered. Each constraint is assigned a unique identifier (DC-XX) for traceability across project artifacts.

2. Constraint Categories Overview

Category	Constraints	Count	Overall Level
Data & Infrastructure	DC-01 through DC-04	4	High
Model & Technical	DC-05 through DC-09	5	High
Clinical & Workflow	DC-10 through DC-13	4	High
Regulatory & Compliance	DC-14 through DC-17	4	High
Organizational & Change Management	DC-18 through DC-20	3	Medium

3. Data & Infrastructure Constraints

DC-01	EEG Acquisition Infrastructure	High
High-density EEG recording equipment capable of capturing NREM sleep data is not universally available across psychiatric clinical settings. The feasibility study used pre-extracted research-grade features — not raw EEG signals from clinical hardware. Deployment requires standardized equipment, calibration protocols, and trained technical staff at every site.		

- Standardized EEG acquisition hardware required at all deployment sites
- Trained EEG technicians and consistent NREM sleep recording protocols needed
- Variability across hardware vendors and acquisition environments creates data quality risk

DC-02	Data Quality & Standardization	High
The model was trained on research-grade, quality-controlled EEG data. Real clinical EEG data introduces additional noise, artifact contamination, and inconsistent acquisition protocols across sites. A standardized data quality and preprocessing pipeline must be established before reliable model inference can be performed. – Clinical EEG data quality varies significantly across settings and hardware – Artifact removal and signal quality validation protocols required – Preprocessing pipeline must be validated against clinical-grade data before deployment		

DC-03	Data Storage & HIPAA Compliance	High
Patient EEG data is sensitive healthcare information subject to HIPAA regulations. Deployment requires encrypted data transmission, secure storage with role-based access controls, Business Associate Agreements (BAAs) with any third-party vendors, and defined data retention and deletion policies. – End-to-end encryption required for all data transmission and storage – Role-based access control and audit logging are mandatory – BAAs required with all third-party cloud or infrastructure vendors		

DC-04	EHR / EMR Integration Complexity	High
Integration with existing Electronic Health Record systems requires dedicated Health IT resources, API development, testing, and validation. No live integration architecture was implemented during the feasibility study. Implementation complexity varies significantly across institutions and EHR platform versions. – HL7 FHIR-based interfaces required for Epic, Cerner, and equivalent platforms – Secure API connectivity, audit logging, and workflow synchronization must be built and tested – Integration timelines and costs vary significantly by institution		

4. Model & Technical Constraints

DC-05	Sample Size & Generalizability	High
The current model was trained on 130 subjects — a sample size insufficient for clinical-grade generalization. Model performance on populations outside the training distribution (different age ranges, ethnicities, comorbidities, and medication profiles) is unknown and unvalidated. – Substantially larger subjects across diverse demographics are recommended before deployment consideration – Demographic coverage must include the full age range, varied ethnicities, and clinical severity levels – Comorbid psychiatric conditions and medication effects on EEG features were not assessed		

DC-06	Model Retraining & Versioning	High
Clinical ML models require ongoing performance monitoring, periodic retraining as new data becomes available, and formal version control. A model deployed without a retraining protocol risks performance degradation over time as patient population characteristics and clinical practices evolve. – Formal model versioning and change management protocol required – Performance monitoring dashboards needed post-deployment – Retraining triggers and approval processes must be defined before go-live		

DC-07	Feature Extraction Pipeline	High
The current study used pre-extracted EEG features provided in the research dataset. A deployable system requires an automated, validated feature extraction pipeline that processes raw EEG signals and generates the same feature set the model was trained on — consistently and reliably across different hardware environments. – Raw EEG to feature extraction pipeline must be built and validated – Feature consistency across hardware vendors must be tested – Pipeline must replicate research-grade preprocessing under clinical conditions		

DC-08	Model Explainability	High
The SVM model used in this study is a black-box classifier. Clinical deployment requires interpretable outputs — psychiatrists need to understand which EEG features drive a risk score before they can responsibly act on it. A formal explainability layer must be implemented before any clinical use. – Feature importance at inference time required for each patient-level prediction – ANOVA-based importance used in feasibility study — deployment requires inference-time explanations – Clinician-readable output format required — not raw feature scores		

DC-09	Inference Reliability & Failover	Medium
In a clinical environment, system failures during patient evaluation create workflow disruption and patient safety risks. A deployment-grade system requires defined failover protocols, error handling, and graceful degradation procedures to ensure clinician workflows are not dependent on system availability. – Failover and recovery procedures must be documented and tested – System errors must be surfaced clearly to clinicians without disrupting workflow – Graceful degradation mode required when model inference is unavailable		

5. Clinical & Workflow Constraints

DC-10	Clinical Validation Requirement	High
-------	---------------------------------	------

The model has not been validated in a real clinical environment. Before deployment, a prospective clinical validation study is required — evaluating model performance on live patient data under real conditions, with outcomes tracked longitudinally across clinical sites.

- Prospective multi-site clinical validation study required
- Physician-supervised pilot studies needed before any broader rollout
- Real-world performance may differ significantly from retrospective research results

DC-11**Diagnostic Risk — False Positives & Negatives****High**

Psychiatric healthcare involves high clinical stakes. The system must be designed with explicit awareness of diagnostic error risks in both directions.

- False positives may increase patient anxiety, trigger unnecessary follow-up, or contribute to diagnostic escalation
- False negatives may delay critical intervention, reduce treatment opportunities, or impact patient safety
- Model performance thresholds must be calibrated with clinical risk tolerance — not just statistical metrics

DC-12**Clinician Training & Adoption****High**

Successful deployment depends on clinician trust, interpretability, and workflow usability. Psychiatrists require structured training on interpreting CDSS outputs, understanding system limitations, and applying appropriate judgment. Without this, misinterpretation of output poses direct patient safety risks.

- Formal training program required for all clinical users of the CDSS
- Workflow usability assessment and clinician feedback loops needed pre-deployment
- Change management strategy required to address potential workflow resistance

DC-13**Patient Consent Framework****High**

Patients must be informed that AI-assisted tools are used in their evaluation. A formal consent framework — including explanation of the system's role, limitations, and the clinician's ultimate diagnostic authority — must be developed and integrated into clinical intake processes.

- Informed consent documentation required before AI-assisted evaluation
- Patient-facing explanation of CDSS role and limitations must be developed
- Consent framework must comply with applicable healthcare ethics and legal standards

6. Regulatory & Compliance Constraints

DC-14**FDA SaMD Classification & Submission****High**

The system would be classified as Software as a Medical Device (SaMD) under FDA guidance. Pre-market submission is required before any commercial or clinical deployment. No predicate device for EEG-based schizophrenia risk classification currently exists — a De Novo pathway may be applicable.

- 510(k) or De Novo pre-market submission required before deployment
- Quality Management System (QMS) and detailed technical documentation required

- No regulatory submission or clinical certification process has been initiated

DC-15	HIPAA Compliance	High
All aspects of the system — data collection, transmission, storage, processing, and output — must comply with HIPAA Privacy and Security Rules. This includes encryption standards, access logging, breach notification protocols, and BAAs with all vendors. – Full HIPAA Privacy and Security Rule compliance required – Breach notification protocols and incident response plans must be defined – All third-party vendors must execute Business Associate Agreements		

DC-16	IRB Approval	High
Any prospective clinical data collection involving human subjects requires Institutional Review Board (IRB) approval. This applies to validation studies, pilot deployments, and any research use of patient data beyond the anonymized feasibility dataset. – IRB approval required before any prospective clinical data collection – Applies to all validation studies, pilot programs, and human subjects' research – IRB protocol must document AI system role, risks, and patient protections		

DC-17	Audit Trail & Documentation	High
Regulatory compliance requires complete, tamper-evident audit logs of all system inputs, model inferences, outputs, and clinician interactions. These logs must be retained per applicable healthcare governance requirements and made available for regulatory review. – Timestamped audit logs required for all data inputs, inferences, and outputs – Logs must be tamper-evident and retained per applicable data governance standards – Audit documentation must be available for FDA and compliance review on demand		

7. Organizational & Change Management Constraints

DC-18	Institutional AI Governance	Medium
Healthcare institutions considering deployment would need to establish an AI governance committee or equivalent oversight body responsible for approving, monitoring, and reviewing AI-assisted clinical tools on an ongoing basis. – Formal AI governance policy required before deployment approval – Ongoing monitoring, review, and escalation procedures must be defined – Governance body must include clinical, technical, compliance, and administrative representation		

DC-19	Interdepartmental Coordination	Medium
-------	--------------------------------	--------

Deployment requires alignment across psychiatry, Health IT, compliance, legal, and administration departments. Each stakeholder group has distinct requirements, timelines, and approval processes that must be coordinated before deployment can proceed.

- Cross-functional deployment working group required
- Stakeholder alignment across clinical, technical, and compliance teams essential
- Conflicting departmental priorities must be resolved through governance processes

DC-20	Cost & Resource Planning	Medium
The full cost of deployment — including EEG infrastructure, EHR integration, staff training, regulatory submission, ongoing maintenance, and model monitoring — has not been estimated. A formal cost-benefit analysis is required before any institutional investment decision can be made. – Full cost-benefit analysis required before investment decision – Costs include: EEG hardware, EHR integration, staff training, regulatory submission, and ongoing maintenance – ROI framework should account for both direct clinical value and indirect risk reduction		

8. Constraint Summary

ID	Constraint	Category	Level
DC-01	EEG acquisition infrastructure	Data & Infrastructure	High
DC-02	Data quality & standardization	Data & Infrastructure	High
DC-03	Data storage & HIPAA compliance	Data & Infrastructure	High
DC-04	EHR / EMR integration complexity	Data & Infrastructure	High
DC-05	Sample size & generalizability	Model & Technical	High
DC-06	Model retraining & versioning	Model & Technical	High
DC-07	Feature extraction pipeline	Model & Technical	High
DC-08	Model explainability	Model & Technical	High
DC-09	Inference reliability & failover	Model & Technical	Medium
DC-10	Clinical validation requirement	Clinical & Workflow	High
DC-11	Diagnostic risk — false positives & negatives	Clinical & Workflow	High
DC-12	Clinician training & adoption	Clinical & Workflow	High
DC-13	Patient consent framework	Clinical & Workflow	High
DC-14	FDA SaMD classification & submission	Regulatory & Compliance	High
DC-15	HIPAA compliance	Regulatory & Compliance	High
DC-16	IRB approval	Regulatory & Compliance	High

ID	Constraint	Category	Level
DC-17	Audit trail & documentation	Regulatory & Compliance	High
DC-18	Institutional AI governance	Organizational	Medium
DC-19	Interdepartmental coordination	Organizational	Medium
DC-20	Cost & resource planning	Organizational	Medium

9. Recommended Next Steps

Before deployment consideration, the following actions are required in priority order:

1. Use substantially larger subjects' datasets across diverse demographics (age, sex, ethnicity, comorbidities)
2. Conduct external validation studies on independent clinical cohorts
3. Build and validate an automated raw EEG feature extraction pipeline
4. Implement an inference-time explainability framework for clinical outputs
5. Perform a formal bias and fairness audit across all demographic subgroups
6. Initiate FDA regulatory consultation and identify 510(k) or De Novo pathway
7. Develop HIPAA-compliant data infrastructure, storage, and transmission architecture
8. Design and execute a prospective clinical validation study under physician supervision
9. Conduct a clinician usability evaluation and develop a training program
10. Establish an institutional AI governance framework and obtain IRB approval

10. Conclusion

This assessment highlights that successful healthcare AI deployment requires significantly more than model accuracy alone. The EEG-based clinical decision support system demonstrates meaningful proof-of-concept results but faces 20 identified deployment constraints across five dimensions.

Each constraint would require significant validation, governance, and operational planning before deployment consideration. However, they collectively confirm that the system is not deployment-ready in its current form. The constraints documented here should be treated as a structured pre-deployment checklist for any future initiative seeking to progress this concept toward clinical implementation.

Real-world implementation depends on the alignment of clinical validation, operational integration, regulatory compliance, ethical governance, and healthcare workflow acceptance — all of which are achievable with appropriate investment, coordination, and time.

Prepared by:

Bhumi Nagar | Business Analyst | January 2026

MBA, Business Analytics (STEM) – Saint Peter's University | B. Pharm — L.M. College of Pharmacy